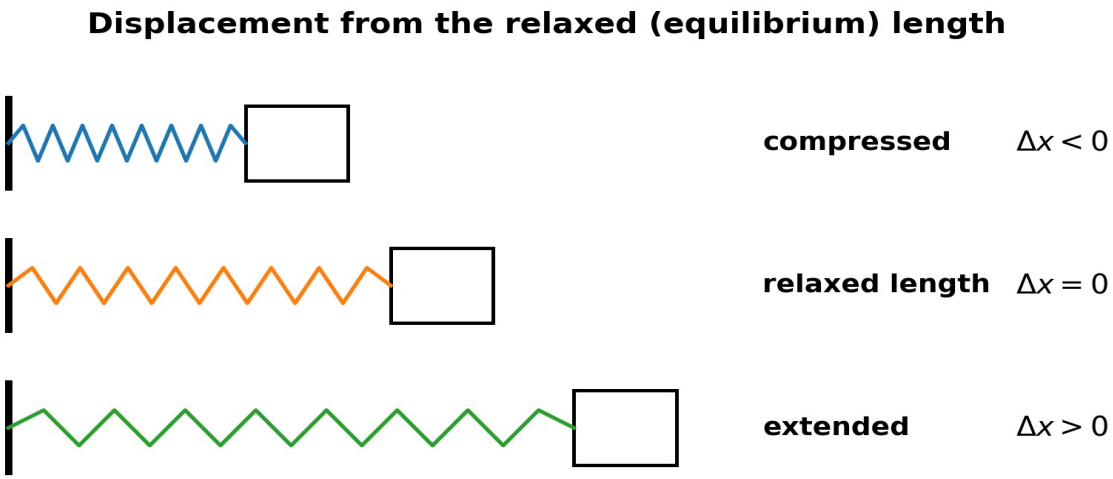
Elastic Potential Energy

AP Physics 1 · Unit 3 · Lesson 4



How to use this pack

Work on the written questions first. Open the accompanying HTML walkthrough when you want the spring experiment, a hint, or a check of your answer. Numerical working should follow: equation → rearrange if needed → substitute SI values → final answer with units.

Learning goals

• Recall the force–displacement relationship for an ideal spring.

• Connect variable-force work to area under a force–displacement graph.

• Derive the elastic potential-energy expression for an ideal spring.

• Calculate elastic energy changes for extension and compression, including changes between two nonzero displacements.

• Use conservation of energy in spring–kinetic and spring–gravitational energy situations.

1. Quick review: ideal springs

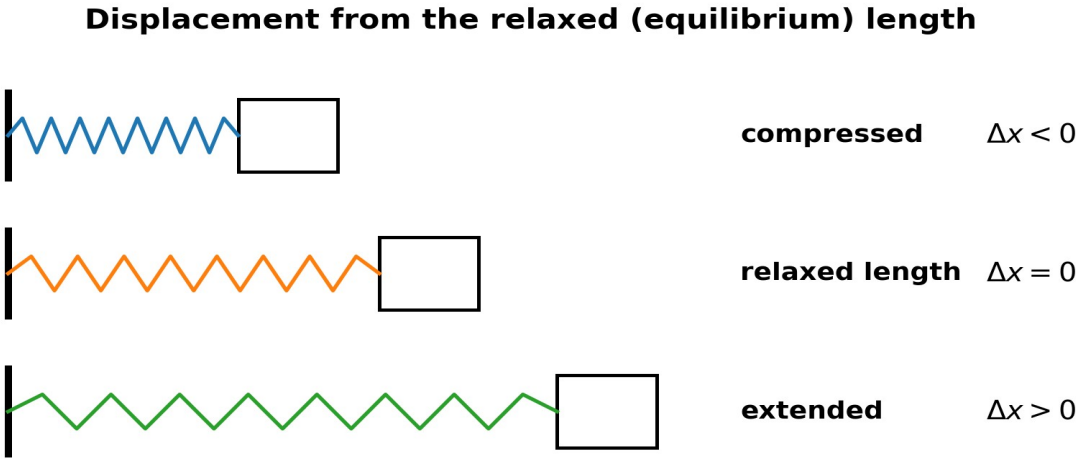
An ideal spring has negligible mass and exerts a restoring force proportional to its displacement from the relaxed (equilibrium) length. The College Board equation is:

$$\vec{F}_s = -k\,\Delta\vec{x}$$

The minus sign is directional: the spring force points toward the equilibrium position. If the spring is extended to the right, the spring force is to the left; if compressed to the left, the spring force is to the right.

Magnitude form used in the graph activity

When a spring is stretched slowly, an external applied force balances the spring force. Its magnitude is $F = k|\Delta x|$. For extension only ($\Delta x > 0$), this is simply $F = k\Delta x$. The slope of an applied-force versus extension graph is therefore k.



Q1. For an ideal spring, what does the spring constant k tell you physically? State its SI unit.

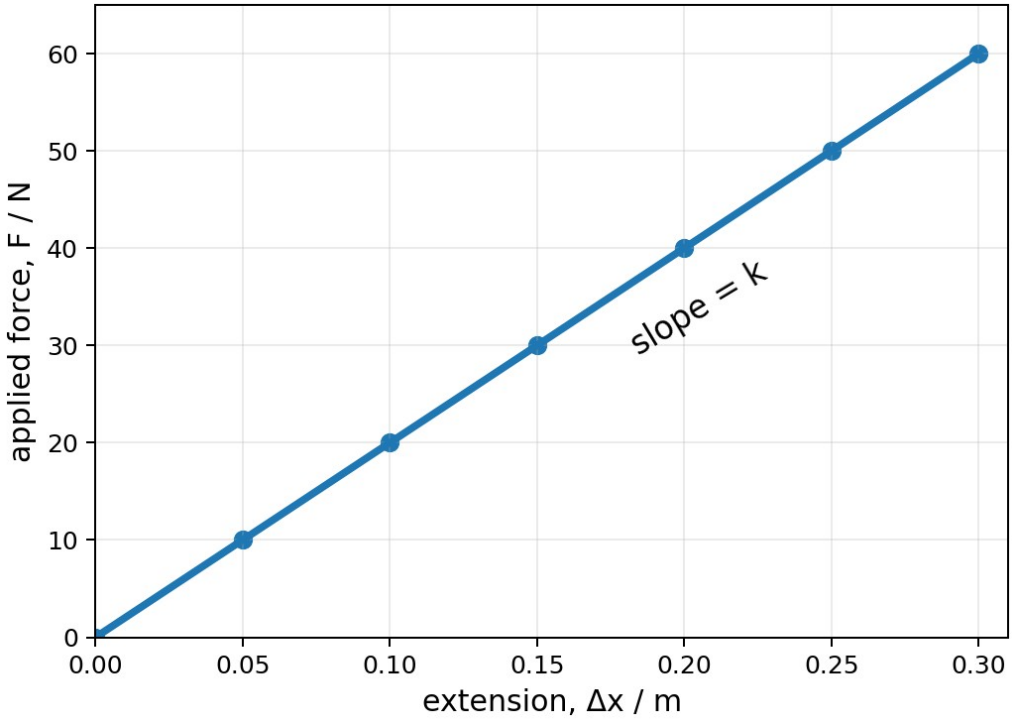
Q2. A spring has k = 180 N m^{-1}. Calculate the magnitude of the spring force when it is extended by 0.065 m.

Q3. State the direction of the spring force for (a) an extension and (b) a compression.

2. Build the force–extension graph

Use the HTML “Spring experiment”. Increase the applied force one step at a time. For each step, record the extension and force in the table below. Do not reveal the best-fit line until you have predicted its shape.

step	applied force F / N	extension Δx / m
0		
1		
2		
3		
4		
5		
6		



Q4. Describe the shape of the graph and state what its gradient represents.

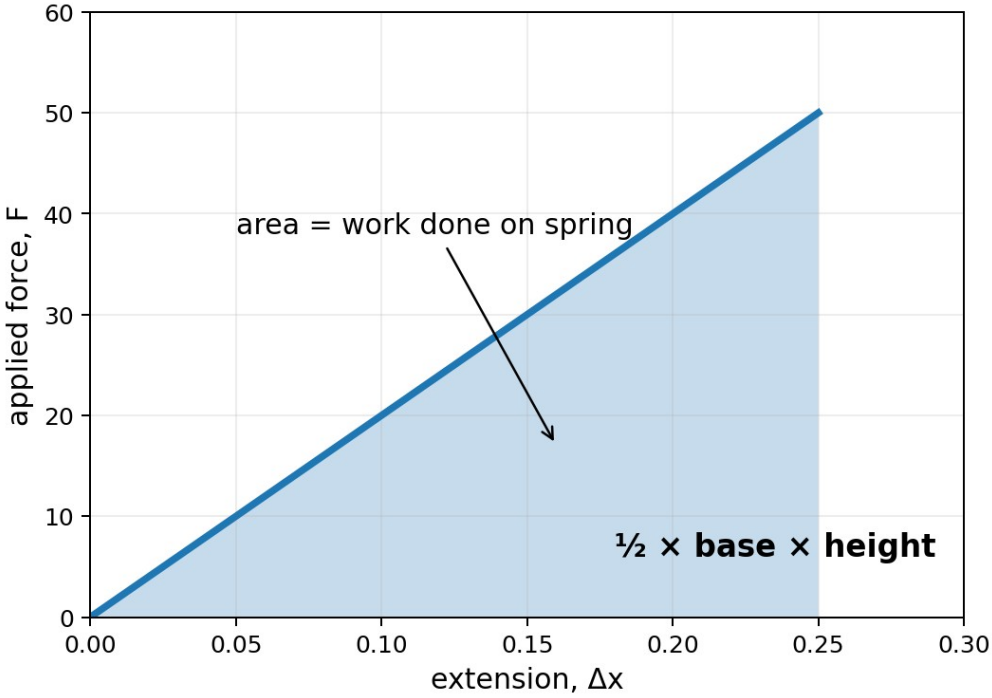
Q5. Use your graph to determine the spring constant k.

3. Variable force and work

For a variable force, work is found from the signed area under a graph of the component of force parallel to the displacement, plotted against displacement.

W = area under the $F_{\parallel} - x$ graph

For the slow stretching experiment, the applied force increases linearly from zero. The area under the applied-force graph is a triangle.



Q6. A spring is slowly stretched from Δx = 0 to Δx. Explain why the work done on the spring is the triangular area under the F–Δx graph.

4. Derivation of elastic potential energy

Stretch the spring slowly from its relaxed length. At each instant the applied force is equal in magnitude and opposite in direction to the spring force, so the applied-force magnitude is F = kΔx for an extension.

Step 1: work done on the spring = area under the applied-force versus extension graph.

Step 2: the area is a triangle with base Δx and height F.

$$W_{\text{on spring}} = \frac{1}{2} F \Delta x = \frac{1}{2} k (\Delta x)^2$$

Step 3: the work done on the spring becomes energy stored reversibly in the ideal spring. Taking $U_s = 0$ at the relaxed length:

$$U_s = \frac{1}{2} k (\Delta x)^2$$

Important
U_s is a scalar and depends on the square of the displacement from the relaxed length. A spring compressed by 0.10 m stores the same elastic potential energy as the same spring extended by 0.10 m.

Q7. Complete the derivation in your own words, beginning with "work = area under the force–displacement graph" and ending with the AP equation for U_s.

5. Using the spring-energy equation

Case A — from the relaxed length to one displacement

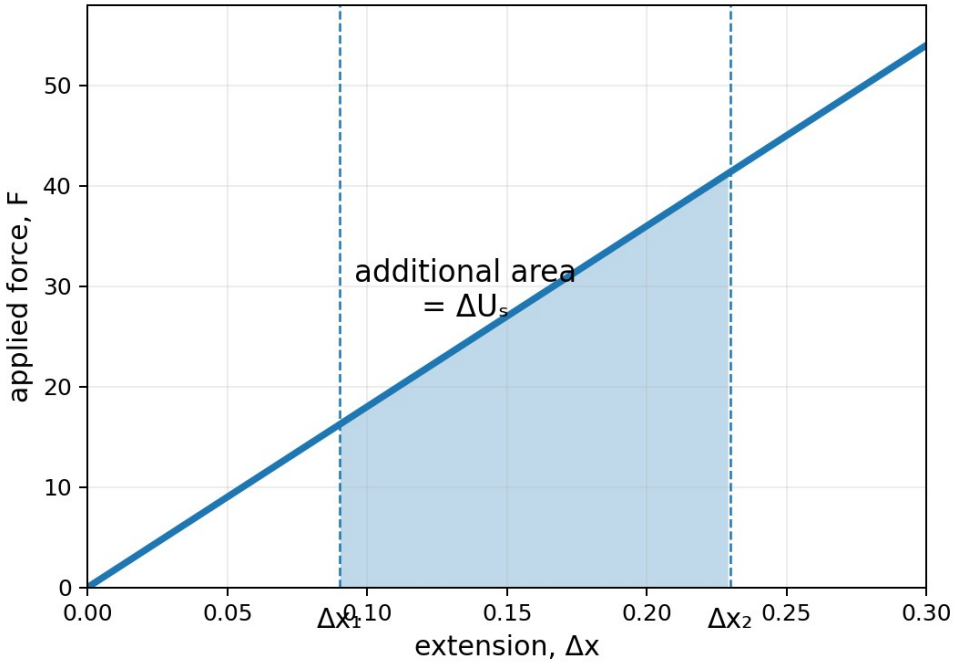
$$U_s = \frac{1}{2} k (\Delta x)^2$$

Q8. A spring with k = 250 N m^{-1} is stretched from its relaxed length to Δx = 0.120 m. Determine the elastic potential energy stored.

Case B — between two nonzero displacements

Do not square the change in extension. Calculate the elastic potential energy at the initial extension and at the final extension, then subtract.

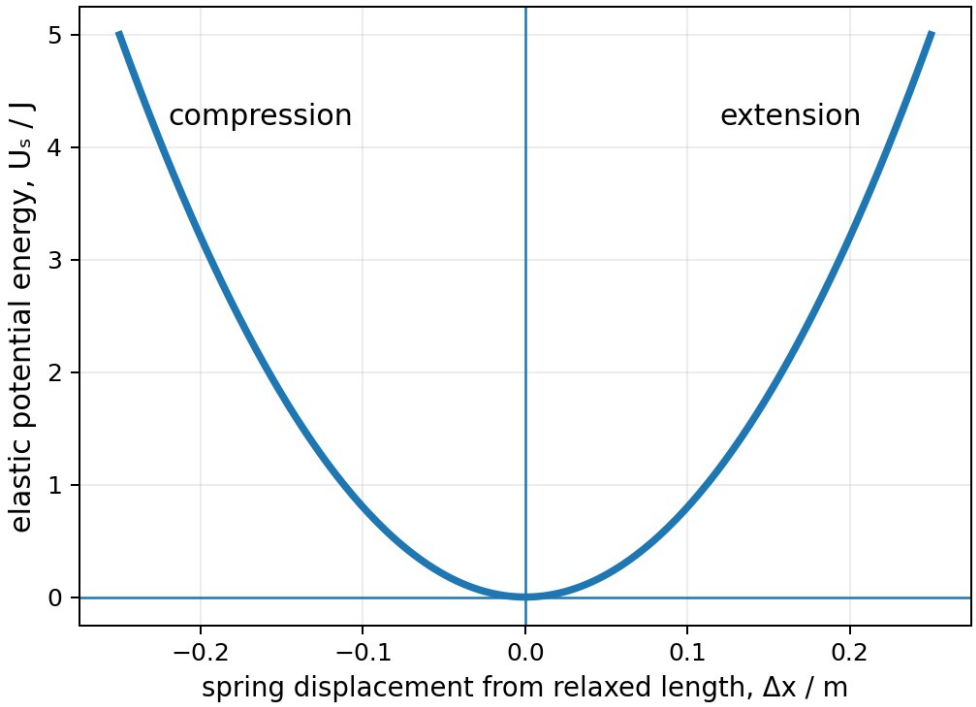
$$\Delta U_s = \tfrac{1}{2}k(\Delta x_f)^2 - \tfrac{1}{2}k(\Delta x_i)^2$$



Q9. A 300 N m^{-1} spring is already extended by 0.080 m and is then stretched to 0.140 m. Determine the increase in elastic potential energy.

Q10. The same spring relaxes from 0.140 m to 0.080 m. Determine ΔU_s. What does the sign tell you?

Case C — compression



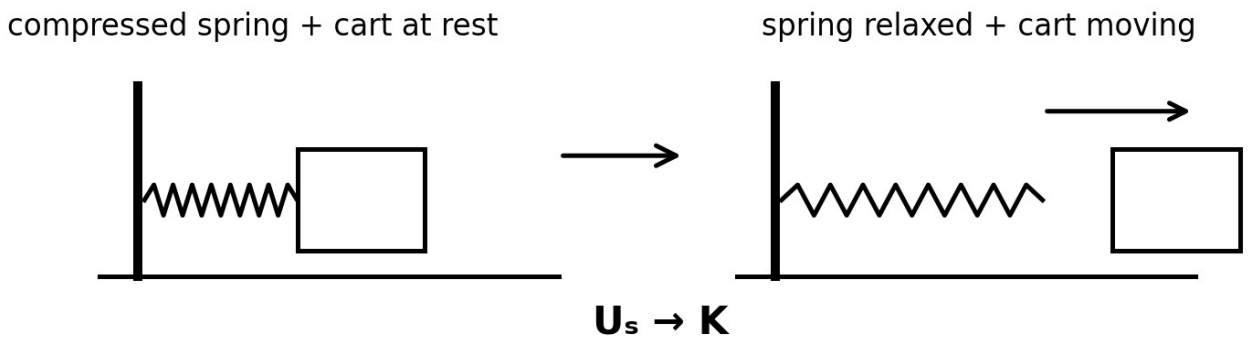
Q11. A spring with k = 400 N m^{-1} is compressed by 0.075 m from its relaxed length. Determine U_s.

6. Energy transfers involving springs

In an ideal system with no dissipative transfer, a decrease in one form of mechanical energy is balanced by an increase in another. Choose the system carefully, identify initial and final states, then write a conservation statement.

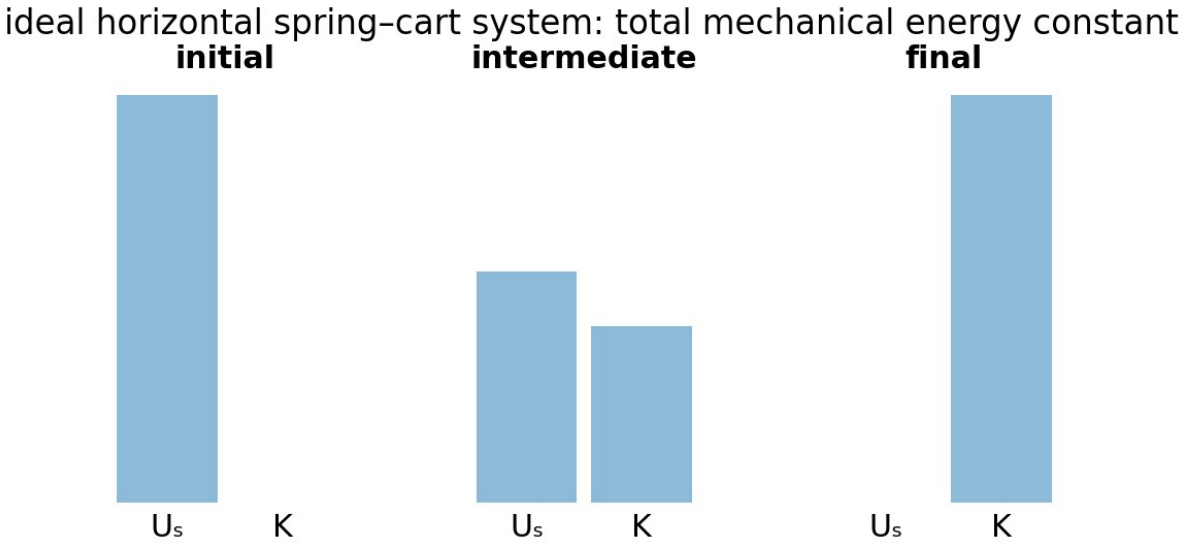
$$\Delta K + \Delta U_s + \Delta U_g = 0$$

A. Spring energy ↔ kinetic energy

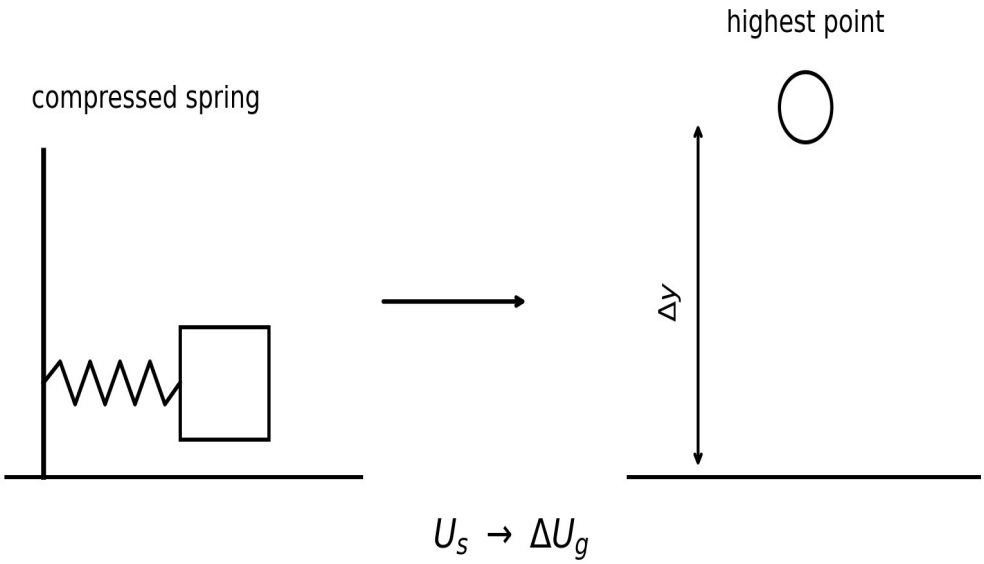


Q12. A 0.40 kg cart is released from rest against a spring of k = 180 N m⁻¹ compressed by 0.150 m on a frictionless horizontal track. Determine the cart speed when the spring reaches its relaxed length.

Q13. A 0.60 kg cart moving at 2.50 m s⁻¹ compresses an ideal spring of k = 250 N m⁻¹ on a frictionless horizontal track. Determine the maximum compression.



B. Spring energy ↔ gravitational potential energy



Q14. A spring launcher has k = 200 N m^{-1} and is compressed by 0.100 m. It launches a 0.250 kg object vertically. Neglect losses. If the object is momentarily at rest at its maximum height and the spring is then relaxed, determine the increase in vertical position from the initial state to the highest point.

Q15. A 1.00 kg object moves downward while stretching a vertical spring from 0.050 m to 0.250 m. The object starts and ends at rest and its vertical displacement is −0.200 m. Neglect losses. Determine k.

7. Summary checklist

☐ I can distinguish the restoring spring force from the external force used to stretch a spring slowly.

☐ I can explain why spring force varies with displacement.

☐ I can obtain work from the signed area under a force-component–displacement graph.

☐ I can derive $U_s = \frac{1}{2}k(\Delta x)^2$.

☐ I can calculate ΔU_s between any two displacements, not only from zero.

☐ I know compression and extension both store positive U_s because displacement is squared.

☐ I can solve spring ↔ kinetic and spring ↔ gravitational energy problems using a clear initial/final energy statement.

AP check

This lesson was checked against AP Physics 1 Topic 2.8 (Spring Forces), Topic 3.2 (Work), Topic 3.3 (Potential Energy), and the current AP Physics 1 Exam Reference Information.